

Erdős–Straus Conjecture

Any fraction with 4 on top can be split into three fractions with 1 on top.

One whole



$4/5$, as four fifths

$4/5$, as three unit fractions



$$1/2 + 1/4 + 1/20 = 10/20 + 5/20 + 1/20 = 16/20 = 4/5$$

Four fifths of a bar, rebuilt from a half, a fourth, and a twentieth. Same length exactly.

TRY IT

$$4/5 = 1/2 + 1/4 + 1/20$$

$$4/7 = 1/2 + 1/15 + 1/210$$

$$4/6 = 1/2 + 1/7 + 1/42$$

WHAT MATHEMATICIANS KNOW

- Paul Erdős and Ernst Straus proposed it in 1948.
- It has been checked by computer for every denominator up to a hundred million billion (10^{17}).
- Writing fractions as sums of unit fractions is ancient. Egyptian scribes did it around 1650 BCE, so these are called Egyptian fractions.

WHY IT'S STILL UNSOLVED

Mathematicians have methods that work for most denominators, including every even one. But a few stubborn families of numbers aren't covered by any known method, and no one has found one approach that always works.

TRY THIS WITH STUDENTS

- Try the greedy method: take away the biggest unit fraction that fits, then repeat. Try $4/5$, then $4/13$. When does it need more than three pieces?
- Ask: If you can split $4/5$, can you split $4/10$ for free? (Double every bottom number.)
- Use fraction strips so students can see the pieces fit exactly.

WORDS TO KNOW

Unit fraction a fraction with 1 on top, like $1/2$ or $1/20$

Denominator the bottom number of a fraction

Online, with the other seven problems: ohiomathteacher.github.io/applet-library/open-problems/erdos-straus.html · The picture was computed from the real numbers.